

Daniele Moretto

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I am a PhD student in Sciences of Civil, Environmental and Architectural Engineering at the University of Padova. My research is focused on numerical methods for coupled multi-physics problems in porous media, reservoir simulation, and the development of general-purpose computational frameworks for subsurface applications.

Education

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| Oct 2023 – Present | PhD in Sciences of Civil, Environmental and Architectural Engineering
<i>Università degli Studi di Padova</i> <ul style="list-style-type: none">• Advisors: Prof. Massimiliano Ferronato and Prof. Andrea Franceschini. |
| May 2025 – Nov 2025 | Visiting Student Researcher
<i>Stanford University – Department of Energy Science and Engineering</i> <ul style="list-style-type: none">• Six-month visiting research period hosted at the Stanford School of Earth, Energy & Environmental Sciences. |
| 2021 – 2023 | Master's Degree in Civil Engineering
<i>Università degli Studi di Padova</i> <ul style="list-style-type: none">• Thesis: <i>GReS – General Reservoir Simulator: development of a general-purpose simulator for coupled multi-physics reservoir problems.</i>• Advisors: Prof. Massimiliano Ferronato, Prof. Andrea Franceschini.• Grade: 110/110 <i>cum laude</i>. |
| 2018 – 2021 | Bachelor's Degree in Civil Engineering
<i>Università degli Studi di Padova</i> <ul style="list-style-type: none">• Thesis: <i>Finite element method for the solution of the mechanical problem in porous media.</i>• Advisor: Prof. Massimiliano Ferronato.• Grade: 110/110 <i>cum laude</i>. |

Research & Teaching Experience

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| Oct 2023 – Present | Doctor of Philosophy (PhD)
<i>Università degli Studi di Padova – DICEA</i> <ul style="list-style-type: none">• Research on numerical algorithms for coupled multiphysics and multidomain simulations in fractured porous media |
| 2024 | Integrated Didactic – Numerical Analysis (laboratory)
<i>Università degli Studi di Padova</i> <ul style="list-style-type: none">• Teaching assistant for undergraduate numerical analysis laboratory. |
| 2023 | Integrated Didactic – Calculus (exercises)
<i>Università degli Studi di Padova</i> <ul style="list-style-type: none">• Teaching assistant for undergraduate calculus course. |

Journal Papers

- Moretto, D., Franceschini, A., Ferronato, M. (2026). A Stabilized Mortar Method for Discontinuities in Geological Media with Non-Conforming Grids. arXiv preprint arXiv:2603.18905.
- Moretto, D., Franceschini, A., Ferronato, M. (2025). A stabilized formulation for the mortar method with non-linear contact constraints. *Applications of Mathematics*, 70(6), 825-849.

- Moretto, D., Franceschini, A., Ferronato, M. (2025). A novel mortar method integration using radial basis functions. *Computers & Mathematics with Applications*, 198, 38-58.

Contributed Talks to Conferences

- *GReS: A Novel Multi-Physics and Multi-Domain Prototyping Platform for Coupled Geomechanics in Porous Media* (with A. Franceschini and M. Ferronato) Siam conference on mathematical & computational issues in the geosciences (GS25). Baton Rouge, Louisiana, October 13-17, 2025.
- *A novel and efficient mortar strategy for multi-physics multi-domain simulations in subsurface applications* (with A. Franceschini and M. Ferronato), 25th Conference on Computational Methods in Water Resources (CMWR) – Tucson, Arizona, October 1–3, 2024.
- *A novel quadrature algorithm for mortar method based on Radial Basis functions* (with A. Franceschini and M. Ferronato), GIMC SIMAI Young 2024 – Naples, Italy, July 10–12, 2024.

Computer Skills

Coding	C++, Python, MATLAB, Bash, L ^A T _E X
Software and other tools	ParaView, Abaqus, Microsoft Office

Languages

Italian	Mother Tongue
English	TOEFL C1 level
Spanish	Basic level